

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396451
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Smalley; John et al.

Pest repellant system with compliant architecture

Abstract

The present disclosure provides a pest repellant system that includes pulse generator circuitry to generate pulses from a power source. The frequency and amplitude of the pulses are selected based on a selected pest species. One or more emitters are coupled to the pulse generator to generate pulsed fields to interfere with navigational/landing/nesting abilities of the selected pest species. The pulse generator circuitry includes filter circuitry to remove/attenuate unwanted frequency components of the pulses, in accordance with regulatory and/or government requirements. The pulse generator circuitry is encased in a housing structure to provide moisture and tamper resistance in accordance with regulatory and/or government requirements.

Inventors:	Smalley; John (Tucson, AZ), Rogers; David (Tucson, AZ)
Applicant:	Symterra, Inc. (Tucson, AZ)
Family ID:	1000008777764
Assignee:	Symterra , Inc. (Tucson, AZ)
Appl. No.:	18/376335
Filed:	October 03, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250107519 A1	Apr. 03, 2025

Publication Classification

Int. Cl.: A01M29/26 (20110101)

U.S. Cl.:

Field of Classification Search**CPC:** A01M (29/26); A01M (29/16); A01M (31/002)**References Cited****U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2006265	12/1934	Davis	N/A	N/A
2223813	12/1939	Brown	N/A	N/A
2647228	12/1952	Just	N/A	N/A
3699976	12/1971	Abe et al.	N/A	N/A
4097838	12/1977	Fiala	N/A	N/A
4366644	12/1982	Lawrence	N/A	N/A
4370534	12/1982	Brandon	N/A	N/A
4802057	12/1988	Patterson et al.	N/A	N/A
4870779	12/1988	Johnson et al.	N/A	N/A
4965552	12/1989	Price et al.	N/A	N/A
H998	12/1990	Gerharz	N/A	N/A
5339564	12/1993	Wilson et al.	N/A	N/A
5468938	12/1994	Roy	N/A	N/A
5473836	12/1994	Liu	N/A	N/A
5575106	12/1995	Martin et al.	N/A	N/A
5774088	12/1997	Kreithen	N/A	N/A
5832657	12/1997	Jan	116/22A	A01M 31/002
5896696	12/1998	Stokes	43/132.1	A01M 1/226
5930946	12/1998	Mah	N/A	N/A
5968401	12/1998	Roy	N/A	N/A
6192622	12/2000	Haj-Yousef	N/A	N/A
6249417	12/2000	Pippen	N/A	N/A
6250255	12/2000	Lenhardt	119/713	A01M 29/16
6996029	12/2005	Boyd et al.	N/A	N/A
7106216	12/2005	Maher	N/A	N/A
7240455	12/2006	Moore	N/A	N/A
7324408	12/2007	Cilliers	N/A	N/A
7698853	12/2009	Ragon et al.	N/A	N/A
7707767	12/2009	Ragon et al.	N/A	N/A
7712247	12/2009	Wijenberg et al.	N/A	N/A
8111164	12/2011	Bryce	N/A	N/A
8733015	12/2013	Lagunas-Solar et al.	N/A	N/A
9185899	12/2014	Richter et al.	N/A	N/A
9450301	12/2015	Cink	N/A	N/A
9581165	12/2016	Babbitt et al.	N/A	N/A
9629354	12/2016	Cohen	N/A	N/A
9775334	12/2016	Dykstra	N/A	N/A
9775337	12/2016	Duncan	N/A	A01M 29/18

10188093	12/2018	Kumar et al.	N/A	N/A
10212795	12/2018	Handelman et al.	N/A	N/A
10694738	12/2019	Garcia	N/A	N/A
10772317	12/2019	Venugopalsetty et al.	N/A	N/A
10820587	12/2019	Stoll	N/A	N/A
11116200	12/2020	Hodam	N/A	N/A
11291198	12/2021	Carnell	N/A	A01M 29/24
11672244	12/2022	Rose et al.	N/A	N/A
11712035	12/2022	Hagen et al.	N/A	N/A
11779007	12/2022	Crisp et al.	N/A	N/A
11957121	12/2023	Tam	N/A	N/A
2004/0200439	12/2003	Thomas et al.	N/A	N/A
2005/0039379	12/2004	Pollinger	N/A	N/A
2006/0024195	12/2005	Lagunas-Solar et al.	N/A	N/A
2011/0023792	12/2010	Osyka	N/A	N/A
2012/0091123	12/2011	Joines	N/A	N/A
2012/0263021	12/2011	Connell	N/A	N/A
2013/0305590	12/2012	Bessei et al.	N/A	N/A
2014/0334268	12/2013	O'Hara	N/A	N/A
2015/0084751	12/2014	Crawford	N/A	N/A
2015/0150237	12/2014	Valls	N/A	N/A
2017/0020122	12/2016	Mirzakhani Nafchi	N/A	N/A
2018/0084774	12/2017	Stoll	N/A	A01M 29/28
2018/0206532	12/2017	Kaczanowski	N/A	N/A
2019/0008131	12/2018	Austin	N/A	A01M 25/002
2019/0313630	12/2018	Stoll	N/A	N/A
2020/0068866	12/2019	Taylor	N/A	N/A
2021/0022332	12/2020	Chong	N/A	N/A
2021/0029983	12/2020	Deering	N/A	A01M 31/002
2021/0259236	12/2020	Garcia et al.	N/A	N/A
2022/0369620	12/2021	Mangler	N/A	N/A
2023/0232811	12/2022	Einat	N/A	N/A
2023/0240281	12/2022	De Samber	119/215	A01M 31/002
2023/0309548	12/2022	Moraes et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2880976	12/2014	EP	N/A
2019222822	12/2018	WO	N/A
2022241005	12/2021	WO	N/A

OTHER PUBLICATIONS

European Search Report and Written Opinion from corresponding Appln. No. EP23201441.5, dated Mar. 5, 2024. 8 pages. cited by applicant

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to pest repellent systems, and, more particularly, to pest repellent systems with regulatory and/or governmental compliant architecture.

BACKGROUND

(2) Bird pest management is a significant problem in a wide variety of commercial, municipal and military concerns. Several products exist on the market for bird abatement and control, for example, spikes, roof coatings, high frequency sonic generators, etc., however, none of the currently available bird abatement approaches provide a lasting and consistent deterrence for birds to land and/or nest in unwanted areas. Birds are known to be disease carriers, and are known to cause significant damage to structures (e.g., buildings, billboards, farm and livestock areas, etc.). Thus, workers assigned for bird cleanup and bird damage repair are often exposed to toxic work environments, and are also often exposed to hazardous work environments from the corrosive effects of bird droppings and urine, the build-up of bird droppings and urine, etc.

(3) As is known for particular bird species of pigeons (members of Columbidae family), low power electromagnetic pulses of a particular frequency (approximately 120 Hz) can interfere with birds' ability to navigate. Such electromagnetic pulses can be used to deter and repel birds from landing and/or nesting on or near structures or areas, for example, buildings, billboards, farm and livestock shelters, airport and airfield areas, etc. However, it is not known in the art the particular frequency needed to deter other varieties of bird species. In addition, it is not known in the art a signal strength needed to deter varieties of bird species. In addition, for pigeon deterrence using pulsed electromagnetic signals at approximately 120 Hz, it is not known if pigeons will eventually adapt to that frequency and return to land and/or nest in unwanted areas.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features and advantages of various embodiments of the claimed subject matter will become apparent as the following Detailed Description proceeds, and upon reference to the Drawings, wherein like numerals designate like parts, and in which:

(2) FIG. 1 illustrates a pest repellent system according to embodiments of the present disclosure;

(3) FIG. 2A illustrates a cross-section of an emitter according to another embodiment of the present disclosure; and

(4) FIG. 2B illustrates a cross-section of an emitter according to another embodiment of the present disclosure;

(5) Although the following Detailed Description will proceed with reference being made to illustrative embodiments, many alternatives, modifications and variations thereof will be apparent to those skilled in the art.

DETAILED DESCRIPTION

(6) FIG. 1 illustrates a pest repellent system **100** according to embodiments of the present disclosure. While the following description is in reference to a pest repellent system specific to birds (and, more specifically, to pigeon species), the teachings of the present disclosure may be applied to different kinds of pests, including, for example, avian pests (birds, bats, flying insects (e.g., bees, wasps, hornets, etc.) and/or ground pests including, for example, rodents, insects, etc. As a general

matter, the teachings of the present disclosure may apply to any pest species that uses innate navigational abilities. The system **100** includes a pulse generator **102** generally configured to generate a frequency-controlled and/or voltage-controlled pulse train **111** from a power source **101**, for example an AC power source **101**. The AC power source **101** may include conventional residential and/or commercial AC power, for example, 110/120 V. AC at 60 Hz, 220 V. AC at 60 Hz, 480 V. AC at 60 Hz, and/or other conventional and/or proprietary AC power source. The following description will be in reference to a conventional 110/120 V. AC power source operating at 60 Hz, however the teachings of the present disclosure can be applied to any AC power source.

(7) The system **100** also includes one or more emitters **112A**, **112B** . . . , **112N** generally configured to generate a respective pulsed electromagnetic signal **116A**, **116B** . . . , **116N** in response to the pulse signal train **111**. The emitters **112A**, **112B** . . . , **112N** may be placed on or near structures, for example, billboards, rooftops, etc., and/or on or near specified areas, for example, fields, entryways, private/commercial/military airfield facilities, etc., so that the pulsed electromagnetic field signal **116A**, **116B** . . . **116N** repel birds away from those structure/areas, as generally illustrated by the flock of birds **118** moving away from the emitters **112A**, **112B** . . . , **112N** in response to the presence of the pulsed electromagnetic field signals **116A**, **116B** . . . , **116N**. The number of emitters **112A**, **112B** . . . **112N** may be selected for a given operating environment and/or to provide coverage for a selected area, such that a sufficient number of emitters **112A**, **112B** . . . , **112N** are selected to generate a sufficient field strength to avoid coverage gaps, thus preventing birds to land/nest in unwanted areas. For example, to deter pigeons from landing/nesting on a billboard structure, the emitters **112A**, **112B** . . . , **112N** may be spaced approximately 3-10 feet apart along the base or catwalk of a billboard. As another example, to deter pigeons from landing/nesting on a rooftop of a building or other structure the emitters **112A**, **112B** . . . **112N** may be spaced approximately 3-10 feet apart along the peak and/or periphery of the rooftop.

(8) The emitters **112A**, **112B** . . . , **112N** are illustrated in cross section in FIG. 1. Using emitter **112A** as an example, the emitter generally includes an insulative body **120A** and a metallic disk **114A** disposed within the insulative body **120A**. The insulative body **120A** may be formed of any suitable non-conductive material such as plastic, polyethylene, silicon, etc. and such material may be selected to have a desired hardness and/or weather resistance and/or UV resistance for a given operating environment. The insulative body **120A** may also include a notch **121A** extending around the body. The metallic disk **114A** may be disposed within the body **120A** and positioned adjacent the notch **121A**, as illustrated. The metallic disk **114A** may be formed of any ferrous and/or conductive material such as for example a metal formed of steel, iron, etc.

(9) The system **100** also includes a conductive wire **126** coupled to the pulse generator **102** and to the emitters **112A**, **112B** . . . **112N**. The conductive wire **126** is selected to have sufficient strength, depending on the length needed to reach all of the emitters **112A**, **112B** . . . , **112N** from the pulse generator **102**, and sufficient diameter to properly conduct relatively high voltages (e.g., 1 kVAC-4 kVAC) of the pulse signal train **111** without significant resistance. In some example embodiments, and again using emitter **112A** as an example, the conductive wire **126** is looped **127** around the body **120A** and disposed within the notch **121A**, i.e., so that the loop of conductive wire **126** surrounds, at least in part, the metallic disk **114A** disposed within the body **120A**. The metallic disk **114A**, in response to the pulses **111** in the wire **126**, operates to generate the pulsed electromagnetic signal **116A**. The emitters **112B** . . . **112N** may be similarly constructed and operate in a similar manner as emitter **112A**, described above.

(10) The pulse generator **102** generally includes controller circuitry **104** generally configured to provide voltage and/or frequency control of the pulse signal **111**, as described below. The pulse generator **102** also includes controllable step-up transformer circuitry **106** coupled to the power source **101** and configured to generate a controllable stepped-up voltage output **107**, for example, in a range of 1000-4000 Volts (V). As a general matter, the controller circuitry **104** is configured to control the controllable transformer circuitry **106** based on user-selectable and/or programmed

voltage control signal (V). To that end, the controllable transformer circuitry **106** may include tap sections to enable controlled changes in the output voltage **107**. In some embodiments, output capacitance and/or rectifier circuitry (not shown) may be coupled to the transformer circuitry **106** to convert the output voltage **107** from AC to DC. Of course, in other embodiments, other known and/or after-developed controllable transformer circuitry may be used, depending on, for example, operating conditions, engineering tolerances, cost considerations, weight considerations, etc.

(11) The pulse generator **102** also includes pulse trigger circuitry **108** generally configured to generate a pulse signals **109**, at a selectable/controllable frequency F, based on the output voltage **107** from the transformer circuitry **106**. The pulse trigger circuitry **108** is generally configured as a momentary ON device, to generate pulse signals **109** having a selected frequency (F) and an amplitude (voltage level) of the output voltage **107** from the transformer circuitry **106**. In some example embodiments, the selected frequency F of the pulse signals **109** is approximately 120 Hz.

(12) The pulse generator **102** also includes radio frequency (RF) filter circuitry **110**, generally configured to remove and/or attenuate unwanted frequency components of the pulse signals **109**, and generate filtered pulse signals **111**. The filter circuitry **110** may include known RF filter circuits, for example, bandpass filter, notch filter, low pass filter, etc. As may be appreciated, certain operating environments (e.g., airports, etc.) and/or government regulatory agencies (e.g., Underwriters Laboratories (UL), Conformité Européenne CE, etc.) may require strict control of RF “bleed” from electronic devices. The filter circuitry **110** may therefore be selected to remove and/or attenuate RF frequency components and/or harmonic content to meet operational and/or governmental requirements.

(13) The pulse generator **102** may also include a housing structure **130** to enclose the electronic components described above. In some embodiments, the housing structure **130** may be a water-tight and/or moisture resistant structure and may be formed from plastic, metal and/or composite materials. The housing structure **130** may fully encapsulate the electronic components to prevent accidental contact. In some embodiments, the housing structure may comply with ingress protection code (e.g., IP65 rated enclosure, IP68 rated enclosure, etc.) defined by the International Electrotechnical Commission (IEC) under the international standard IEC 60529. The housing structure **130** may be a unitary construction (i.e., no openings, no access to internal components) or the housing structure **130** may include a removable portion (e.g., removable lid) to allow access to internal components. In some embodiments, RF/electromagnetic shielding **132** may be disposed (in whole or in part) within the housing structure **130** to reduce and/or attenuate unwanted electromagnetic signals from escaping the housing structure **130** during operation of the pulse generator **102**. Such shielding **132** may include, for example, a metallic lining disposed adjacent to the electronic components, and/or other shielding material as would be understood by one skilled in the art.

(14) The housing structure **130** may also include water-tight seals for each ingress point on the housing structure **130**. For example, a water-tight seal **134A** may be used to form a water-tight opening for a power cable associated with the power source **101**. Similarly, a water-tight seal **134B** may be used to form a water-tight opening for the conductive wire **126**. In some embodiments, the pulse generator **102** may include one or more user interfaces **136** to enable a user to input operational commands. For example, user interface **136** may include buttons, knobs, etc. to enable a user to adjust voltage parameters (V) and or frequency parameters (F). Such user interfaces **136** may also be embodied as water-tight sealed mechanisms to prevent moisture from entering the housing **130**.

(15) In one example embodiment, the controller circuitry **104** may generate a voltage control signal V based on a user-specified voltage. For example, the user-specified voltage V may be derived from a user interface (**136**) associated with the pulse generator **102**. The user interface **136** may include, for example, a knob that a user can rotate to select a desired voltage output **107** of the transformer **106**. In other embodiments, the user interface **136** may be embodied as a display that

can be accessed by a user to select a desired voltage output **107**. The controller circuitry **104** causes the transformer circuitry **106** to generate the output voltage **107** having a voltage level based on the user-supplied input V. Controlling the output voltage **107** also controls the voltage of the pulse signals **111**, and thus controls the signal strength of the pulsed electromagnetic field signals **116A**, **116B . . .**, **116N**.

(16) In another example embodiment, the controller circuitry **104** may generate a voltage control signal based on a pest specific voltage. For example, pulse signals **111** having voltage (amplitude) value of between 1 kV and 4 kV can generate electromagnetic pulses of sufficient strength to deter certain pigeon species away from the emitters **112A**, **112B . . .**, **112N**. However, other pigeon species, and/or other bird species, may require a signal strength that is specific within this range and/or greater than (or less than) 1 kV-4 kV to be a deterrence. Accordingly, controller circuitry **104** may be programmed and/or controlled (e.g., externally controlled) to generate a specific voltage control signal V so that the transformer circuitry **106** generates a specified output voltage **107**. In other words, the controller circuitry **104** may be “tuned” to a specific pest type, thus enhancing the pest repellent ability of the system **100**. By selecting a voltage (or voltage range) for a specific pest type, the teachings of the present disclosure may offer enhanced pest deterrence for targeted pests, while avoiding interference with other animals.

(17) In another example embodiment, the controller circuitry **104** may generate a variable voltage control signal V so that the voltage of operation randomly varies within a selected random voltage range (“Random”). For example, and again using the pigeon example, while the specific voltage to deter some pigeon species is known to be in the range of 1 kV to 4 kV, deterrence of these pigeon species may occur by varying the voltage within this range. Thus, the controller circuitry **104** may generate a variable and random voltage control signal V to cause the output voltage **107** of the transformer **106** to have a random voltage within the range of 1 kV to 4 kV. Of course, this is only an example of the range of voltage operations, and in other embodiments other voltage ranges may be selected. Moreover, in some embodiments, the controller **104** may generate a variable and random voltage control signal V at fixed and/or random intervals. For example, the variable and random voltage control signal V may be generated at user-defined intervals (e.g., every 5 seconds, etc.) or at random intervals within a user-definable range (e.g., 1-20 seconds). It will be appreciated that some bird species may adapt to a fixed operating voltage, thus enabling the birds to land/nest in unwanted areas despite the presence of the pulsed electromagnetic signals **116A**, **116B . . .**, **116N**. By providing randomness in both voltage and timing, birds may not be able to adapt to such conditions and instead permanently seek other areas to land/nest.

(18) In still another example embodiment, the controller circuitry **104** may generate a variable voltage control signal V so that the voltage of operation incrementally changes within a selected range. Thus, for example, the controller circuitry **104** may generate a variable voltage control signal V to cause the transformer circuitry **106** to generate the output voltage **107** having a selected step value within the range of 1 kV to 4 kV. For example, a step value of 10 may be selected so that the output voltage **107** takes on voltage values incremented and/or decremented by 10 V (resulting in increments of 1 kV, 1010 V, 1020 V, and so on). Of course, this is only an example of a step value and range of voltage operations, and in other embodiments other step values and voltage ranges may be selected. Moreover, in some embodiments, the stepped voltage may be generated at fixed and/or random intervals (timing). For example, the stepped voltage may be generated at user-defined intervals (e.g., every 5 seconds, etc.) or at random intervals within a user-definable range (e.g., 1-20 seconds). It will be appreciated that some bird species may adapt to a fixed operating voltage, thus enabling the birds to land/nest in unwanted areas despite the presence of the pulsed electromagnetic signals **116A**, **116B . . .**, **116N**. By providing stepped voltage values at selected and/or random intervals, birds may not be able to adapt to such conditions and instead permanently seek other areas to land/nest.

(19) In another example embodiment, the controller circuitry **104** may generate a frequency control

signal F based on a user-specified frequency. For example, the user-specified frequency F may be derived from a user interface **136** associated with the pulse generator **102**. The user interface **136** may include, for example, a knob that a user can rotate to select a desired frequency of the pulse signals **111**. In other embodiments, the user interface **136** may be embodied as a display that can be accessed by a user to select a desired frequency. The controller circuitry **104** causes the pulse trigger circuitry **108** to generate the pulse signals **111** having a frequency based on the user-supplied frequency input F. Controlling the frequency of the pulse signals **111** also controls the frequency of the pulsed electromagnetic field signals **116A**, **116B** . . . , **116N**.

(20) In other embodiments, and similar to the voltage control described above, the controller circuitry may control the frequency of operation based on pest-specific criteria and/or random as frequency control (within a selected frequency range). By selecting a frequency (or frequency range) for a specific pest type, the teachings of the present disclosure may offer enhanced pest deterrence for targeted pests while avoiding interference with other animals. By providing randomness in both frequency and timing, birds may not be able to adapt to such conditions and instead permanently seek other areas to land/nest.

(21) FIG. 2A illustrates a cross-section of an emitter **112'** according to another embodiment of the present disclosure. In this embodiment, the loop of conducting wire **126** is formed internally within the body **120'** of the emitter **112'**. Connector **238A** is provided to couple the conductive wire **126** to the internal loop **127'** of wire surrounding the disk **114'**. Similarly, connector **238B** is provided to couple the internal loop **127'** of wire surrounding the disk **114'** to a next segment of conductive wire leading to the next emitter in the series (as shown in FIG. 1). Connectors **238A** and **238B** may include, for example, water-tight snap-fit connectors, screw-type connectors, etc. In some embodiments, connector **238B** may be omitted in the case of the emitter being the last in the signal chain. As illustrated, the body **120'** of the emitter **112'** may have generally rectangular cross-section. In other embodiments, the cross-sectional shape of the emitter **112'** may include, for example, round-top, disk-shaped, square, and/or other shapes that may be selected for space and/or aesthetics of a given installation environment. In addition, and with reference to FIG. 1, the emitters may be embodied as preformed emitter strings with an emitter at selected intervals (e.g., 6 ft.) along the length of a continuous conductive wire (thus eliminating the need for connectors **238A** and **238B**). In addition, the disk **114'** is illustrated as having a generally rectangular cross section. In other embodiments, the shape and/or dimension of the disk **114'** may be selected to provide optimal electromagnetic cooperation with the pulses in the loop **127'**.

(22) FIG. 2B illustrates a cross-section of an emitter **112''** according to another embodiment of the present disclosure. In this embodiment, the conductive wire **226** is embodied as an insulated cable, having the conductive wire **126''** surrounded by an insulative jacket **228**. Similar to the previous embodiment, the loop **127'** of conducting wire **126''** is formed internally within the body **120''** of the emitter **112''**. Connector **240A** is provided to couple the conductive wire **126''** to the internal loop **127'** of wire surrounding the disk **114''**. Similarly, connector **240B** is provided to couple the internal loop **127'** of wire surrounding the disk **114''** to a next segment of conductive wire leading to the next emitter in the series (as shown in FIG. 1). Connectors **240A** and **240B** may include, for example, water-tight snap-fit connectors, screw-type connectors, etc. In some embodiments, connector **240B** may be omitted in the case of the emitter being the last in the signal chain. As illustrated, the body **120''** of the emitter **112''** may have generally rectangular cross-section. In other embodiments, the cross-sectional shape of the emitter **112''** may include, for example, round-top, disk-shaped, square, and/or other shapes that may be selected for space and/or aesthetics of a given installation environment. In addition, and with reference to FIG. 1, the emitters may be embodied as preformed emitter strings with an emitter at selected intervals (e.g., 6 ft.) along the length of a continuous conductive wire (thus eliminating the need for connectors **238A** and **238B**). In addition, the disk **114''** is illustrated as having a generally rectangular cross section. In other embodiments, the shape and/or dimension of the disk **114''** may be selected to provide optimal electromagnetic

cooperation with the pulses in the loop 127".

(23) As used in this application and in the claims, a list of items joined by the term "and/or" can mean any combination of the listed items. For example, the phrase "A, B and/or C" can mean A; B; C; A and B; A and C; B and C; or A, B and C. As used in this application and in the claims, a list of items joined by the term "at least one of" can mean any combination of the listed terms. For example, the phrases "at least one of A, B or C" can mean A; B; C; A and B; A and C; B and C; or A, B and C.

(24) "Circuitry", as used in any embodiment herein, may comprise, for example, singly or in any combination, hardwired circuitry, programmable circuitry such as processors comprising one or more individual instruction processing cores, state machine circuitry, and/or firmware that stores instructions executed by programmable circuitry and/or future computing circuitry including hardware embodiments of accelerators such as neural net processors and non-silicon implementations of the above. The circuitry may, collectively or individually, be embodied as components that forms part of a larger system, for example, an integrated circuit (IC), system on-chip (SoC), application-specific integrated circuit (ASIC), programmable logic devices (PLD), digital signal processors (DSP), field programmable gate array (FPGA), logic gates, registers, semiconductor device, chips, microchips, chip sets, etc.

(25) Any of the operations described herein may be implemented in a system that includes one or more non-transitory storage devices having stored therein, individually or in combination, instructions that when executed by circuitry perform the operations. The storage device includes any type of tangible medium, for example, any type of disk including hard disks, floppy disks, optical disks, compact disk read-only memories (CD-ROMs), compact disk rewritables (CD-RWs), and magneto-optical disks, semiconductor devices such as read-only memories (ROMs), random access memories (RAMs) such as dynamic and static RAMs, erasable programmable read-only memories (EPROMs), electrically erasable programmable read-only memories (EEPROMs), flash memories, Solid State Disks (SSDs), embedded multimedia cards (eMMCs), secure digital input/output (SDIO) cards, magnetic or optical cards, or any type of media suitable for storing electronic instructions. Other embodiments may be implemented as software executed by a programmable control device. Also, it is intended that operations described herein may be distributed across a plurality of physical devices, such as processing structures at more than one different physical location.

(26) The terms and expressions which have been employed herein are used as terms of description and not of limitation, and there is no intention, in the use of such terms and expressions, of excluding any equivalents of the features shown and described (or portions thereof), and it is recognized that various modifications are possible within the scope of the claims. Accordingly, the claims are intended to cover all such equivalents. Various features, aspects, and embodiments have been described herein. The features, aspects, and embodiments are susceptible to combination with one another as well as to variation and modification, as will be understood by those having skill in the art. The present disclosure should, therefore, be considered to encompass such combinations, variations, and modifications.

(27) Reference throughout this specification to "one embodiment" or "an embodiment" means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases "in one embodiment" or "in an embodiment" in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

Claims

1. A pest repellant system, comprising: a water-resistant housing structure; transformer circuitry, disposed within the water-resistant housing structure, to generate a step-up voltage output from voltage source; pulse trigger circuitry, disposed within the water-resistant housing structure, to generate pulse signals having a selected frequency based on the voltage output of the transformer circuitry; filter circuitry, disposed within the water-resistant housing structure, to filter signal components in the pulse signals; and controller circuitry, disposed within the water-resistant housing structure, to control the transformer circuitry to generate the voltage output having a voltage level based on, at least in part, a selected species of pest; the controller circuitry also to control the pulse trigger circuitry to control the frequency of the pulse signals based on, at least in part, based on the selected species of pest.
2. The system of claim 1, further comprising: an electromagnetic emitter to receive the plurality of pulses and to generate a pulsed electromagnetic signal having frequency based on the frequency of the plurality of pulses, wherein the frequency of the pulsed electromagnetic signal operates to repel the selected pest species.
3. The system of claim 1, wherein the housing structure is a water-resistant housing structure in compliance with an IP65 or IP68 standard.
4. The system of claim 1, wherein the filter circuitry being selected from a notch filter, a bandpass filter, or a low pass filter, and wherein the filter circuitry to attenuate frequency components in the pulses in compliance with a regulatory and/or governmental restriction on impermissible frequency components in the pulses.
5. The system of claim 1, wherein the housing structure further includes radio frequency (RF) shielding disposed within the housing, the RF shielding to attenuate frequency components in the pulses in compliance with a regulatory and/or governmental restriction on impermissible frequency components in the pulses.
6. The system of claim 1, wherein the housing structure further includes at least one connector to provide power input or output from the housing, the connector being a water-resistant connector in compliance with a regulatory and/or governmental requirements for water resistance.
7. A pest repellant system, comprising: a water-resistant housing structure; transformer circuitry, disposed within the water-resistant housing structure, to generate a step-up voltage output from voltage source; pulse trigger circuitry, disposed within the water-resistant housing structure, to generate pulse signals having a selected frequency based on the voltage output of the transformer circuitry; filter circuitry, disposed within the water-resistant housing structure, to filter signal components in the pulse signals; controller circuitry, disposed within the water-resistant housing structure, to control the transformer circuitry to generate the voltage output having a voltage level based on, at least in part, a selected species of pest; the controller circuitry also to control the pulse trigger circuitry to control the frequency of the pulse signals based on, at least in part, based on the selected species of pest; and an electromagnetic emitter to receive the plurality of pulses and to generate a pulsed electromagnetic signal having frequency based on the frequency of the plurality of pulses, wherein the frequency of the pulsed electromagnetic signal operates to repel the selected pest species.
8. The system of claim 7, wherein the housing structure is a water-resistant housing structure in compliance with an IP65 or IP68 standard.
9. The system of claim 7, wherein the filter circuitry being selected from a notch filter, a bandpass filter, or a low pass filter, and wherein the filter circuitry to attenuate frequency components in the pulses in compliance with a regulatory and/or governmental restriction on impermissible frequency components in the pulses.
10. The system of claim 7, wherein the housing structure further includes radio frequency (RF) shielding disposed within the housing, the RF shielding to attenuate frequency components in the pulses in compliance with a regulatory and/or governmental restriction on impermissible frequency

components in the pulses.

11. The system of claim 7, wherein the housing structure further includes at least one connector to provide power input or output from the housing, the connector being a water-resistant connector in compliance with a regulatory and/or governmental requirements for water resistance.
 12. The system of claim 7, wherein the emitter comprising: a body portion; a metallic member disposed within the body portion; and a loop of conducting wire disposed around, at least in part, the metallic member; wherein the loop of conducting wire being coupled to the pulses and wherein the pulses interact with the metallic member to generate the pulsed electromagnetic signal.
 13. A pest repellent system, comprising: a water-resistant housing structure; transformer circuitry, disposed within the water-resistant housing structure, to generate a step-up voltage output from voltage source; pulse trigger circuitry, disposed within the water-resistant housing structure, to generate pulse signals having a selected frequency based on the voltage output of the transformer circuitry; filter circuitry, disposed within the water-resistant housing structure, to filter signal components in the pulse signals and generate filtered pulse signals; controller circuitry, disposed within the water-resistant housing structure, to control the transformer circuitry to generate the voltage output having a voltage level based on, at least in part, a selected species of pest; the controller circuitry also to control the pulse trigger circuitry to control the frequency of the pulse signals based on, at least in part, based on the selected species of pest; and an electromagnetic emitter to receive the plurality of pulses and to generate a pulsed electromagnetic signal having frequency based on the frequency of the plurality of pulses, wherein the frequency of the pulsed electromagnetic signal operates to repel the selected pest species; wherein the emitter comprising: a body portion; a metallic member disposed within the body portion; and a loop of conducting wire disposed around, at least in part, the metallic member; wherein the loop of conducting wire being coupled to the pulses and wherein the pulses interact with the metallic member to generate the pulsed electromagnetic signal.
 14. The system of claim 13, wherein the housing structure is a water-resistant housing structure in compliance with an IP65 or IP68 standard.
 15. The system of claim 13, wherein the filter circuitry being selected from a notch filter, a bandpass filter, or a low pass filter, and wherein the filter circuitry to attenuate frequency components in the pulses in compliance with a regulatory and/or governmental restriction on impermissible frequency components in the pulses.
 16. The system of claim 13, wherein the housing structure further includes radio frequency (RF) shielding disposed within the housing, the RF shielding to attenuate frequency components in the pulses in compliance with a regulatory and/or governmental restriction on impermissible frequency components in the pulses.
 17. The system of claim 13, wherein the housing structure further includes at least one connector to provide power input or output from the housing, the connector being a water-resistant connector in compliance with a regulatory and/or governmental requirements for water resistance.
 18. The system of claim 13, further comprising a conductive wire coupled to the filtered pulsed signals and to the loop of conductive wire.
 19. The system of claim 18, wherein the conductive wire is an insulated cable.
-